Xiwei Zhao

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Education

- **Aalto University** — Helsinki, Finland
  Master of Science in Robotics and machine learning; GPA: 3.94 (4.79/5.00) — *Aug. 2023 – Present*
- **Harbin Institute of Technology** — Harbin, China
  Bachelor of Engineering in Automation; GPA: 3.61 (90.2/100.0) — *Aug. 2019 – July. 2023*

Publication

- **Video-to-BT: Generating Reactive Behavior Trees from Human Demonstration Videos**
  ***Xiwei Zhao**, Yiwei Wang, Yansong Wu, Fan Wu, et al. Submitted to ICRA 2026. [arXiv] [Website]*

Experience

- **MIRMI, Technical University of Munich** — Munich, Germany
  Master thesis worker — *July. 2025 - Present*
  - **VLM-Driven Reactive Behavior Tree Generation**: Architected a VLM-based system to translate demonstration videos into structured behavior trees for high-level task monitoring.
  - **Force-Position Hybrid Control for Assembly Tasks**: Implemented adaptive impedance control algorithms enabling < 0.5mm positional error in plug-in operations through dynamic torque/position coupling.
- **Seed Robotics, ByteDance** — Beijing, China
  Embodied AI Research Intern — *Jan. 2025 - June. 2025*
  - **Enhanced VLM Decision-Making**: Fine-tuned Qwen-2.5VL-7B to improve VLM reasoning in novel environments, boosting task success rate from 40% to 70% and achieving performance on par with GPT-4o.
  - **Multi-Agent Robotic Planning Framework**: Developed a collaborative multi-agent framework based on LangChain for automated robotic task planning, achieving: 90%+ accuracy in high-level assembly procedure generation aligned with manual specifications; 80% reliability in producing fine-grained manipulation commands
- **Intelligent Robotics Group, Aalto University** — Helsinki, Finland
  Research assistant — *June. 2024 - Dec. 2024*
  - **VLA Models Utilization in Deformable Items Grasping**: Fine-tuned Octo and OpenVLA models for deformable object grasping tasks on IsaacSim. Built a high-precision simulation environment, adapting the models to handle fabrics and plastic bags.
  - **Automatic Dataset Collection in IsaacSim**: Created a plastic bag manipulation framework with Isaac Cortex for automated dataset collection. Implemented behavior tree technique for adaptive subtask transitions, adjusting strategies based on object states.
- **Wireless Lab, Institut national de la recherche scientifique (INRS)** — Montreal, Canada
  Machine Learning Research Intern, Mitacs-CSC scholarship — *July. 2022 - Dec. 2022*
  - **Machine Learning for Channel Estimation**: Optimized channel models in communication systems and applied machine learning algorithms, specifically the Expectation-Maximization (EM) algorithm, to estimate channel transmission parameters.
  - **Mathematical Modeling**: Developed and refined mathematical models to enhance the accuracy and efficiency of channel estimation in communication systems.

Projects

- **QuantSoftware Toolkit**: https://github.com/xiweizhao118/TS_factor_analyst
  Open source python library for financial data analysis and machine learning for finance. Offered an integrated analysis frame with factor calculation and evaluation. Codes are available on GitHub.
- **GPT-based Franka Robot**: https://github.com/xiweizhao118/LLM_Robotics
  Based on the GPT-3.5 Turbo model fine-tuning, enabled Franka to comprehend natural language instructions from users. Conducted simulation using ROS and provided complete code on GitHub repository.

Programming Skills

- **Languages**: Python, C++, JavaScript, Matlab
- **Tools**: VS Code, Git, Latex